A SCHEME HAS A FINITE AFFINE STRATIFICATION EXACTLY WHEN IT IS QUASI-COMPACT AND QUASI-SEPARATED

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Editorial note. Counted span is the author's own proof environment, source lines 21520-21566, immediately following the statement at 21511-21518.

Editorial note. The proof ends by adding one extra stratum Z' as a minimal element of the partially ordered index set; that construction is displayed inside the span.

Editorial note. Four cross-chapter pointers (Coherent, Morphisms, Schemes) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Characterisations of quasi-separated and separated schemes. Schemes, lemmas schemes-lemma-characterize-quasi-separated and schemes-lemma-characterize-separated, cited at proof lines 21524-21530. Turns the intersection behaviour of affine opens into quasi-separatedness, giving the necessity direction.
- Affineness of a morphism is local on the target. Morphisms, lemma morphisms-lemma-characterize-affine, cited at proof lines 21559-21561. Lets the affineness of the pieces U_i cap Z and U_i cap W_j be checked over the affine opens U and V.
- Induction principle for quasi-compact quasi-separated schemes. Coherent, lemma coherent-lemma-induction-principle, cited in the proof. Provides the induction step that builds the finite affine stratification for the sufficiency direction.

Source reference. Original reference 08DR: focused-seeds/geometry/S9-morph-qc-affine-stratification/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01S8: focused-seeds/geometry/S9-morph-qc-affine-stratification/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01KO: focused-seeds/geometry/S9-morph-qc-affine-stratification/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01KP: focused-seeds/geometry/S9-morph-qc-affine-stratification/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a scheme. The following are equivalent*

(1) *X has a finite affine stratification, and*
(2) *X is quasi-compact and quasi-separated.*

Proof. Let $X = \bigcup X_i$ be a finite affine stratification. Since each X_i is affine hence quasi-compact, we conclude that X is quasi-compact. Let $U, V \subset X$ be affine open. Then $U \cap X_i$ and $V \cap X_i$ are affine open in X_i since $X_i \to X$ is an affine morphism. Hence $U \cap V \cap X_i$ is an affine open of the affine scheme X_i (see Schemes, Lemma 01KP for example). Therefore $U \cap V = \coprod U \cap V \cap X_i$ is quasi-compact as a finite union of affine strata. We conclude that X is quasi-separated by Schemes, Lemma 01KO.

Assume X is quasi-compact and quasi-separated. We may use the induction principle of Cohomology of Schemes, Lemma 08DR to prove the assertion that X has a finite affine stratification. If X is empty, then it has an empty affine stratification. If X is nonempty affine then it has an affine stratification with one stratum. Next, assume $X = U \cup V$ where U is quasi-compact open, V is affine open, and we have a finite affine stratifications $U = \bigcup_{i \in I} U_i$ and $U \cap V = \coprod_{j \in J} W_j$. Denote $Z = X \setminus V$ and $Z' = X \setminus U$. Note that Z is closed in U and Z' is closed in V. Observe that $U_i \cap Z$ and $U_i \cap W_j = U_i \times_U W_j$ are affine schemes affine over U. (Hints: use that $U_i \times_U W_j \to W_j$ is affine as a base change of $U_i \to U$, hence $U_i \cap W_j$ is affine, hence $U_i \cap W_j \to U_i$ is affine, hence $U_i \cap W_j \to U$ is affine.) It follows that

$$U = \coprod\nolimits_{i \in I} (U_i \cap Z) \amalg \coprod\nolimits_{(i,j) \in I \times J} (U_i \cap W_j)$$

is a finite affine stratification with partial ordering on $I \amalg I \times J$ given by $i' \leq (i, j) \Leftrightarrow i' \leq i$ and $(i', j') \leq (i, j) \Leftrightarrow i' \leq i$ and $j' \leq j$. Observe that $(U_i \cap Z) \times_X V = \emptyset$ and $(U_i \cap W_j) \times_X V = U_i \cap W_j$ are affine. Hence the morphisms $U_i \cap Z \to X$ and $U_i \cap W_j \to X$ are affine because we can check affineness of a morphism locally on the target (Morphisms, Lemma 01S8) and we have affineness over both U and V. To finish the proof we take the stratification above and we add one additional stratum, namely Z', whose index we add as a minimal element to the partially ordered set. □